The Non-Closure Theorem of FFGFT Torus Modes

Why Products of Different r_i Are Structurally Meaningless

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Preliminary note

The operations discussed below are **algebraically correct**. The restrictions are not arithmetic rules but **consistency conditions of the FFGFT model**: products of different r_i are algebraically defined but structurally meaningless — they correspond to no state in the model.

.1 Prefactors as Geometric Invariants

In FFGFT, for every particle i:

$$m_i = r_i \cdot \xi_0^{p_i} \cdot v, \qquad r_i = f(\mathcal{T}_i), \quad p_i = g(\mathcal{T}_i) \tag{1}$$

where $\mathcal{T}_i = (n_i, l_i, j_i)$ is the torus quantum number of the particle.

The functions f and g are **injective** on the set of admissible torus modes: from (r_i, p_i) one can uniquely reconstruct $\mathcal{T}_i$.

The prefactors r_i are **not free parameters** but **geometric invariants** of the respective particle — fully determined by its quantum numbers (n, l, j).

For electron and muon (Doc. 006):

$$m_e = \frac{4}{3} \cdot \xi_0^{3/2} \cdot v \tag{2}$$

$$m_\mu = \frac{16}{5} \cdot \xi_0^{1} \cdot v \tag{3}$$

.2 The Key Insight: The Set of Pairs (r_i, p_i) Is Not Closed Under Multiplication

Torus quantisation

The admissible torus modes generate r_i and p_i through specific geometric rules (Doc. 006). The admissible pairs (r, p) are discrete and finite:

Particle	(n, l, j)	r_i	p_i
Electron	$(1, 0, 12)$	$4/3$	$3/2$
Muon	$(2, 1, 12)$	$16/5$	1
Tau	$(3, 2, 12)$	$25/9$	$2/3$

The non-closure theorem

Forming the product $r_e \cdot r_\mu$:

$$r_e \cdot r_\mu = \frac{4}{3} \cdot \frac{16}{5} = \frac{64}{15}, \qquad p_e + p_\mu = \frac{3}{2} + 1 = \frac{5}{2}$$

The model asks:

Is there a particle X with $r_X = 64/15$ and $p_X = 5/2$?

Answer: No. In the quantum number table, $64/15$ appears for no p, and $p = 5/2$ appears for no r.

This is not coincidence. The set of admissible pairs $\mathscr{P}$ is **not closed** under multiplication:

$$\forall\, i \neq j \;:\; (r_i \cdot r_j,\; p_i + p_j) \notin \mathscr{P}$$

where $\mathscr{P}$ is the set of all pairs occurring in FFGFT.

Why is this so?

The admissible torus modes have separate, non-multiplicative generation rules for r:

$$r(n,l) = \frac{(l+1)^2}{n^2 - l(l+1)} \cdot \text{(spin factor)}$$

(simplified). The product of two such fractions is **never** again a fraction of the same form with two quantum numbers.

The r_i are therefore not numbers in a field, but **labels of a discrete geometric space with its own composition** — and this composition is not field multiplication.

.3 Algebraically Correct but Structurally Meaningless

The cross-product problem

Inserting (2) and (3) symbolically into $E_0 = \sqrt{m_e \cdot m_\mu}$:

$$E_0^2 = m_e \cdot m_\mu = \underbrace{\frac{4}{3} \cdot \frac{16}{5}}_{r_e \cdot r_\mu} \cdot \xi_0^{3/2+1} \cdot v^2 = \frac{64}{15} \cdot \xi_0^{5/2} \cdot v^2 \tag{4}$$

The result is **algebraically correct**. But because $(64/15,\ 5/2) \notin \mathscr{P}$, this expression corresponds to **no FFGFT particle** — it is structurally uninterpretable.

Analogy: different algebraic spaces

Each particle i lives in the state space spanned by its torus mode $\mathscr{T}_i$. The r_i are coefficients in *different spaces*. Their product corresponds to a tensor product:

$$r_e \cdot r_\mu \;\leftrightarrow\; \mathbf{v}_e \otimes \mathbf{v}_\mu$$

This yields a tensor-product state — **not** a state in the original single-particle space. In a single-particle theory (mass formulas), such a state is meaningless.

Further substitution worsens the result

Inserting (4) into $\alpha = \xi_0 \cdot (E_0/\text{MeV})^2$:

$$\alpha \overset{\text{structurally wrong}}{=} \frac{64}{15} \cdot \xi_0^{7/2} \cdot \frac{v^2}{\text{MeV}^2} \tag{5}$$

Exponent $7/2$ and factor $64/15$ appear nowhere in the basic equations. Additionally K_{frak} is missing and units are covertly mixed. In practice: instead of $-1.35\,\%$, one gets $-3.15\,\%$ deviation.

.4 What Is Permitted and Why

Numerical evaluation: transition to $\mathbb{R}$

Numbers live in $\mathbb{R}$, where multiplication is always permitted. Once r_e and r_μ have been evaluated to numbers, they have **no memory** of their origin from different particles:

$$m_e \approx 0.511\,\text{MeV}, \quad m_\mu \approx 105.66\,\text{MeV}$$
$$E_0 = \sqrt{0.511 \cdot 105.66}\,\text{MeV} = 7.348\,\text{MeV}$$
$$\alpha = \xi_0 \cdot (7.348)^2 \approx 0.00720 \quad (-1.35\,\%)$$

Division: respecting the separate spaces

In ratios, v and K_{frak} cancel, and the quantum number structures *remain separate*:

$$\frac{m_\mu}{m_e} = \frac{r_\mu}{r_e} \cdot \xi_0^{p_\mu - p_e} = \frac{12}{5} \cdot \xi_0^{-1/2} \approx 207.9 \quad (0.5\,\%)$$

The result $12/5$ is interpretable — no cross-product. Division **respects** the separate quantum number spaces.

Basic rule — The non-closure condition

The set of admissible pairs (r_i, p_i) is not closed under multiplication:

$$\boxed{\forall\, i \neq j \;:\; (r_i \cdot r_j,\; p_i + p_j) \notin \mathscr{P}}$$

Therefore: products of different r_i are algebraically defined but correspond to no particle in the model. Only numerical evaluation (transition to $\mathbb{R}$) permits free combination.

.5 Connection to Other Results

The same principle applies to the Koide relation (Doc. 192): there too, symbolic insertion of mass terms generates cross-terms with exponents belonging to no particle.

The rule is not arbitrary, but a **necessary consequence of torus quantisation** and the injectivity of f and g.

Summary

The prefactors r_i are not free parameters but geometric invariants of torus modes with fixed quantum numbers. The set of admissible pairs (r_i, p_i) is not closed under multiplication — a product $r_i \cdot r_j$ with $i \neq j$ is algebraically defined but belongs to no torus mode and is therefore meaningless in the model. Combinations are only permitted after numerical evaluation in $\mathbb{R}$.

Legend

Symbol	Meaning
$\mathscr{T}_i = (n_i, l_i, j_i)$	Torus mode: principal, orbital, spin quantum number
$r_i = f(\mathscr{T}_i)$	Geometric prefactor (injective)
$p_i = g(\mathscr{T}_i)$	Exponent in the mass formula
$\mathscr{P}$	Set of admissible pairs (r_i, p_i)
$v = 246$ GeV	Higgs vacuum expectation value
$\xi_0 = 4/30000$	Geometric base parameter
K_{frak}	Fractal correction (≈ 0.986)
$\mathbb{R}$	Real numbers (no structural memory)
$\mathbf{v}_e \otimes \mathbf{v}_\mu$	Tensor-product (meaningless in single-particle theory)
$(64/15,\ 5/2)$	Cross-pair $\notin \mathscr{P}$, no FFGFT particle

References: Doc. 006 (quantum numbers), Doc. 043, Doc. 011, Doc. 044 (α), Doc. 192 (Koide, general principle).